

Exercise - Week 6: MATH1061/7861

Duration: 30 minutes

2 Questions - 10 marks

Topics: Euclidean Algorithm, Sequences, Mathematical Induction,
Strong Mathematical Induction and the Well-Ordering Principle

Student Name:

Student number: |_|_|_|_|_|_|_|_|

*Your student number should be 8 digits and **not** start with an s. e.g. 41234567*

(1) **(4 marks)**

(a) **(2 marks)** Use the Euclidean Algorithm to compute $\gcd(252, 198)$.

(b) **(2 marks)** Find the prime factorisation of 252 and 198.

(2) **(6 marks)** The Fibonacci Sequence $F_1, F_2, F_3, \dots$ is defined by

$$F_1 = 1, \quad F_2 = 1, \quad F_{n+1} = F_n + F_{n-1} \quad \forall n \geq 2.$$

Use strong induction to prove that $F_n < 2^n$ for all positive integers n .

(1 mark) Base cases:

(2 marks) Inductive assumption:

(1 mark) What we need to prove:

(2 marks) Inductive step: